



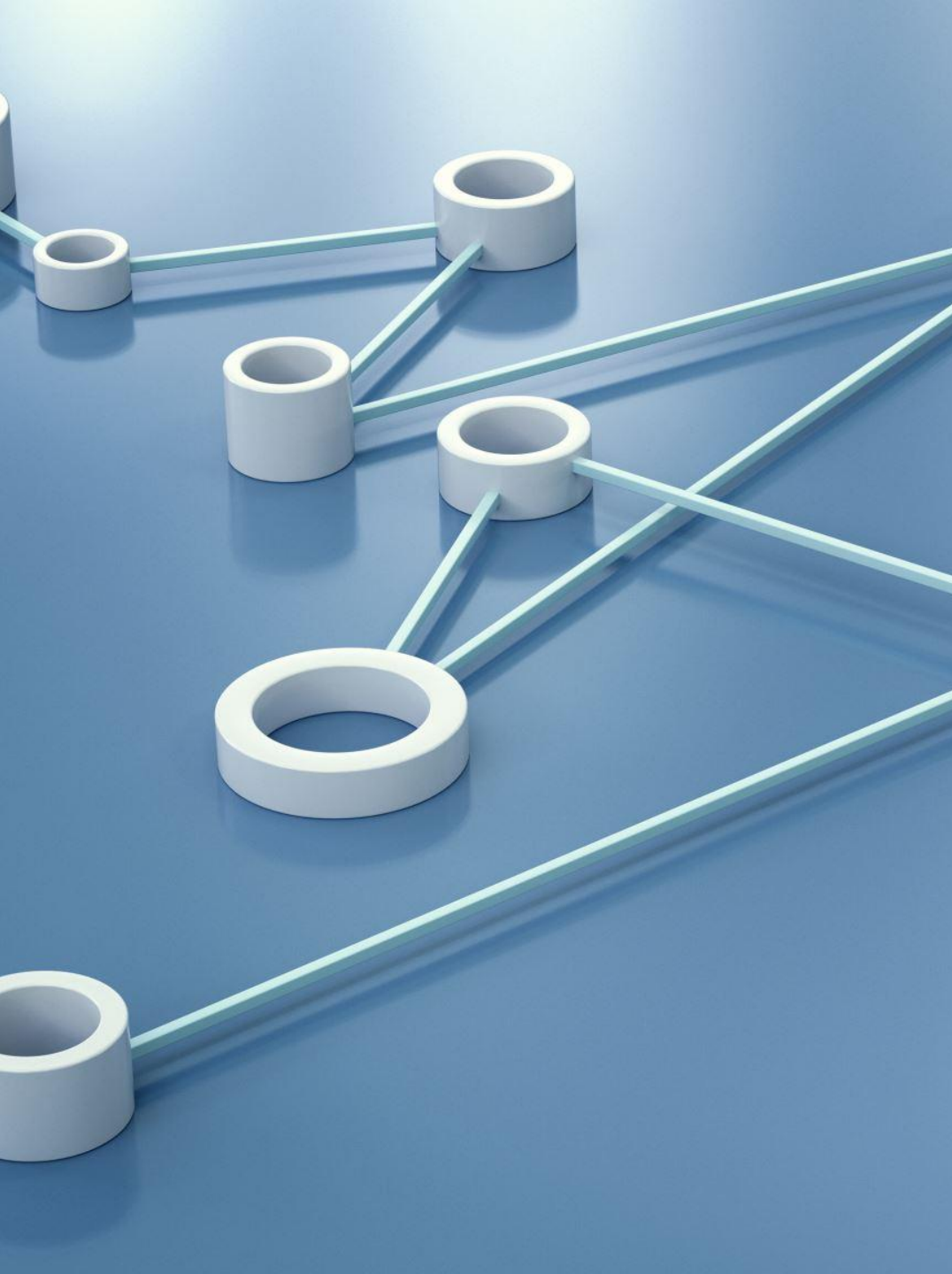
Database Fundamentals

An Overview of Databases

Objectives

- Understand the fundamental concepts of databases.
- Discuss the history of Databases
- Identify the job opportunities related to Database Management
- Understand the importance of databases to Management Information Systems (MIS)





The need to learn the fundamentals of databases

- To understand how data is stored, organized, and retrieved.
- To design efficient systems that avoid redundancy and errors.
- To write accurate queries for extracting meaningful information.
- To ensure data integrity, security, and consistency.
- To apply database concepts in real-world applications (banking, e-commerce, healthcare, etc.).
- To build a strong foundation for advanced topics like big data, machine learning, and cloud computing.

The Importance of Data

- Data is one of the most critical assets of any business.
- It is used and collected practically everywhere, from businesses trying to determine consumer patterns based on credit card usage, to space agencies trying to collect data from other planets.



What companies do with stored data

1

Analyze customer behavior and market trends.

2

Improve products, services, and operations.

3

Personalize user experiences.

4

Detect fraud, errors, or risks.

5

Generate reports for management and regulators.

6

Support strategic planning and innovation.

Why data need to be stored



To keep records for future reference.



To ensure business continuity and compliance.



To track transactions, activities, and history.



To enable analysis and decision-making.

The Importance of Data

- Data, as important as it is, needs robust, secure, and highly available software that can store and process it quickly.
- The answer to these requirements is a ***solid and a reliable database***.

What is a database?

- ***A database is a data repository designed to support efficient data storage, retrieval, and maintenance.***
- Multiple types of databases exist to suit various industry requirements.
- A database may be specialized to store binary files, documents, images, videos, relational, multidimensional, transactional, analytic, or geographic data.

What is a database?

- Data can be stored in various forms, namely tabular, hierarchical and graphical forms.
- If data is stored in a ***tabular form***, then it is called a ***relational database***.
- When data is organized in a ***tree structure form***, it is called a ***hierarchical database***.
- ***Data stored as graphs*** representing relationships between objects is referred to as a ***network database***.

What is a database management system (DBMS)?

- While a database is a repository of data,

A database management system, or simply DBMS, is a set of software tools that control access, organize, store, manage, retrieve, and maintain data in a database.

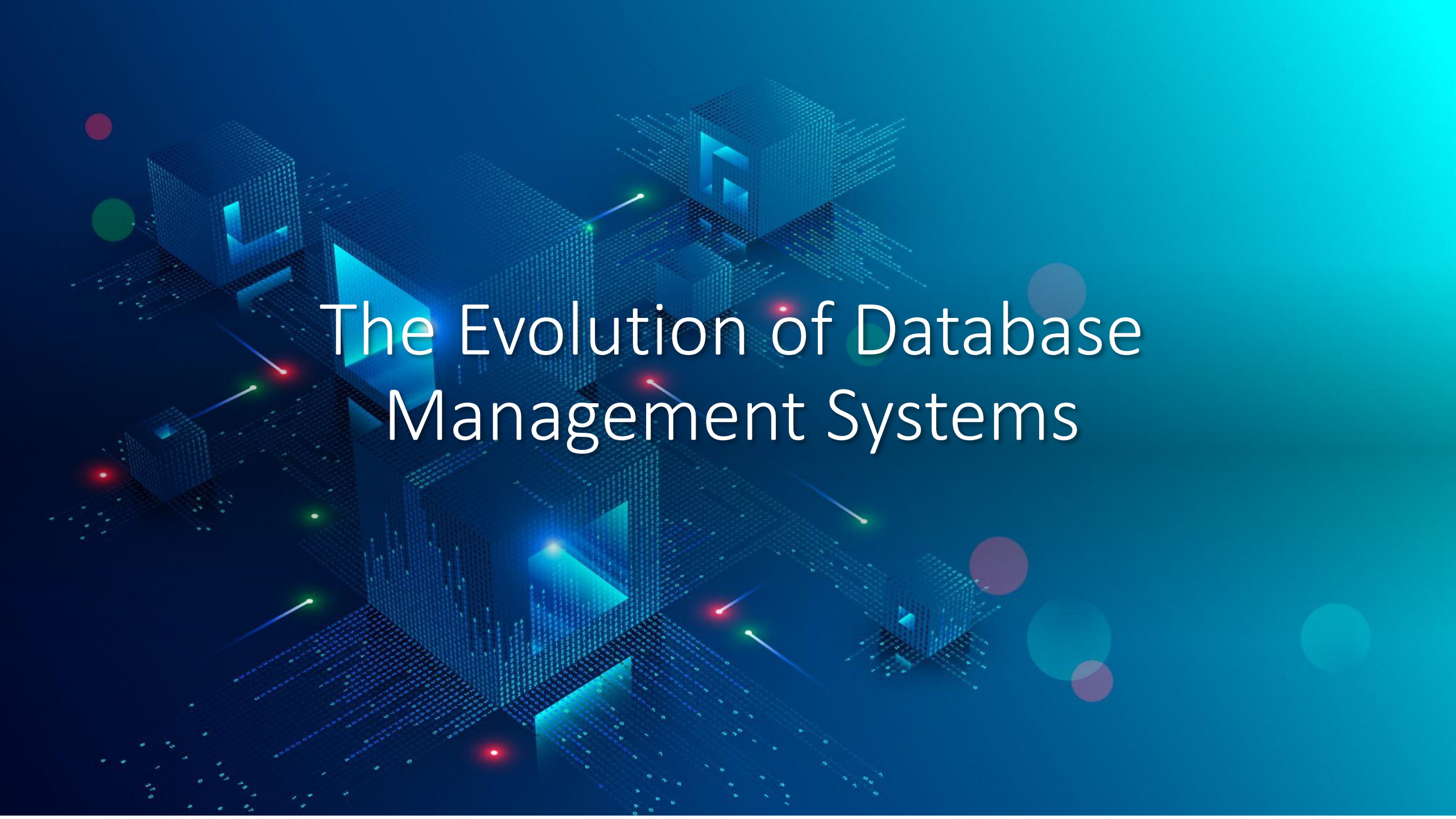
Why do we need database software or a DBMS? Can we not just store data in simple text files for example?

- The answer lies in the way users access the data and the handle of corresponding challenges.
- First, we need the ability to have multiple users insert, update and delete data to the same data file without "stepping on each other's toes".

Why do we need database software or a DBMS?
Can we not just store data in simple text files for example?

- *We need to have a standard interface for data access, tools for data backup, data restore, and recovery, and a way to handle other challenges, such as the capability to work with huge volumes of data and users.*
- Database software has been designed to handle all of these challenges.

- The most mature database systems in production are relational database management systems (RDBMS's).
- RDBMS's serve as the backbone of applications in many industries including banking, transportation, health, etc.

The background is a deep blue gradient. It features several 3D cubes made of binary code (0s and 1s). Some cubes have glowing blue rectangular openings. Thin, colorful lines (red, green, blue) streak across the scene, and small glowing dots in red, green, and blue are scattered throughout. On the right side, there are larger, semi-transparent circles in shades of blue and purple.

The Evolution of Database Management Systems

1960's

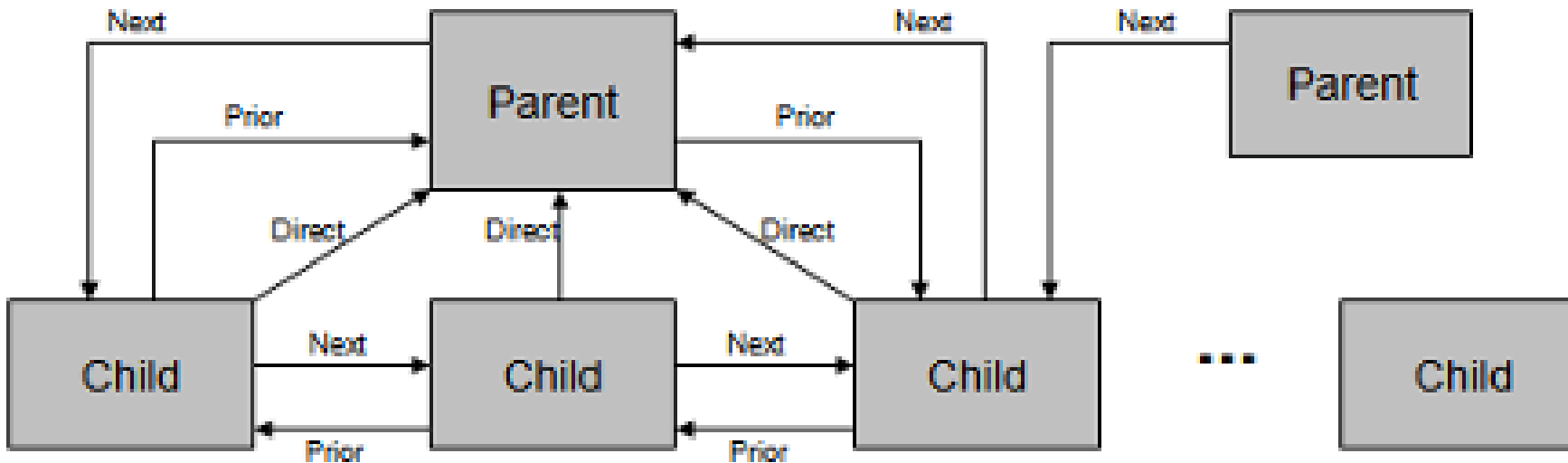
- Network and hierarchical systems like **CODASYL**, **IMS** were the leading technologies, supporting banking, accounting, and order processing on commercial mainframes.
- Their design tightly coupled physical and logical data handling, so when data was moved on disk, applications had to be modified to match the new locations.

1970's

- In 1970, **E.F. Codd** of IBM introduced the **relational model** in his paper *“A Relational Model of Data for Large Shared Data Banks.”* This concept brought **data independence**, separating physical storage from logical representation.
- Data could be moved from one part of the disk to another or stored in a different format without causing applications to be rewritten.

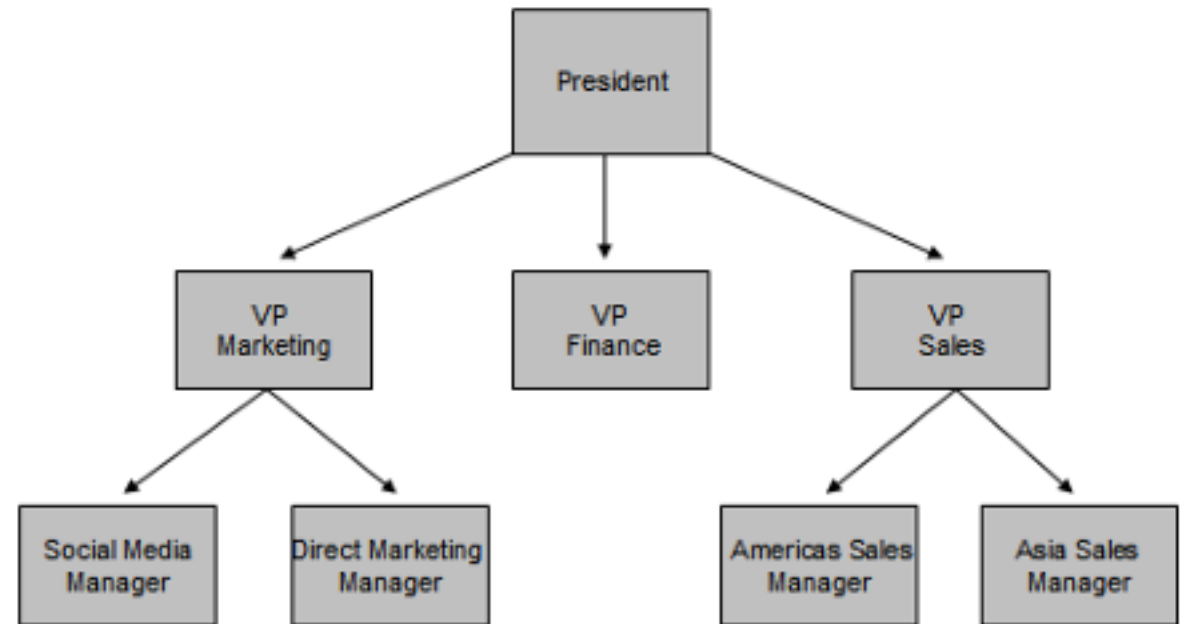
The Network Model

- In 1969, CODASYL (Committee on Data Systems Languages) released its first specification about the network data model.



The Hierarchical Model

- The hierarchical model organizes its data using a tree structure. The root of the tree is the parent followed by child nodes.
- A child node cannot have more than one parent, though a parent can have many child nodes.



The image shows a server room with multiple rows of server racks. The racks are filled with various server units, some of which have indicator lights glowing. The background is out of focus, showing more racks and a bokeh effect of yellow and blue lights. The text 'The Database Management System' is overlaid on the left side of the image.

The Database Management System

What is a DBMS?

A database management system (DBMS) is a collection of programs that enables users to create and maintain a database.

The ***DBMS*** is hence a general-purpose software system that facilitates the processes of defining, constructing, manipulating, and sharing databases among various users and applications.

What is a DBMS?

- ***Defining*** a database involves specifying the data types, structures, and constraints for the data to be stored in the database.
- ***Constructing*** the database is the process of storing the data itself on some storage medium that is controlled by the DBMS.
- ***Manipulating*** a database includes such functions as querying the database to retrieve specific data, updating the database to reflect changes in the miniworld, and generating reports from the data.
- ***Sharing*** a database allows multiple users and programs to access the database concurrently.

What is a DBMS?

- ***Protection*** includes both system protection against hardware or software malfunction (or crashes), and security protection against unauthorized or malicious access.
- A typical large database may have a life cycle of many years, so the DBMS must be able to maintain the database system by allowing the system to evolve as requirements change over time.

A Simplified Database System Environment

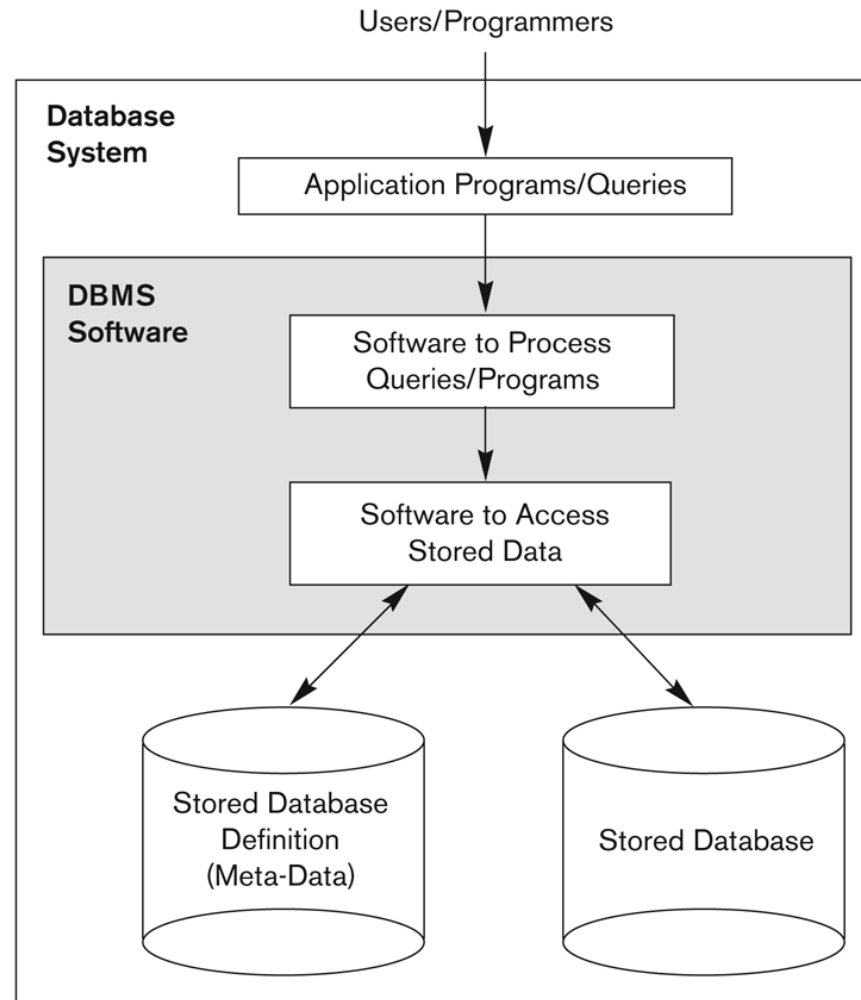


Figure 1.1
A simplified database
system environment.

Example

STUDENT	Name	StudentNumber	Class	Major
	Smith	17	1	CS
	Brown	8	2	CS

COURSE	CourseName	CourseNumber	CreditHours	Department
	Intro to Computer Science	CS1310	4	CS
	Data Structures	CS3320	4	CS
	Discrete Mathematics	MATH2410	3	MATH
	Database	CS3380	3	CS

SECTION	SectionIdentifier	CourseNumber	Semester	Year	Instructor
	85	MATH2410	Fall	98	King
	92	CS1310	Fall	98	Anderson
	102	CS3320	Spring	99	Knuth
	112	MATH2410	Fall	99	Chang
	119	CS1310	Fall	99	Anderson
	135	CS3380	Fall	99	Stone

GRADE_REPORT	StudentNumber	SectionIdentifier	Grade
	17	112	B
	17	119	C
	8	85	A
	8	92	A
	8	102	B
	8	135	A

PREREQUISITE	CourseNumber	PrerequisiteNumber
	CS3380	CS3320
	CS3380	MATH2410
	CS3320	CS1310

FIGURE 1.2 A database that stores student and course information.

Example

- A UNIVERSITY database for maintaining information concerning students, courses, and grades in a university environment. Figure 1.2 shows the database structure and a few sample data for such a database.
- The database is organized as five files, each of which stores data records of the same type.
- The STUDENT file stores data on each student, the COURSE file stores data on each course, the SECTION file stores data on each section of a course, the GRADE_REPORT file stores the grades that students receive in the various sections they have completed, and the PREREQUISITE file stores the prerequisites of each course.

Example

- To define this database, we must specify the structure of the records of each file by specifying the different types of data elements to be stored in each record.
- In Figure 1.2, each STUDENT record includes data to represent the student's Name, StudentNumber, Class (freshman or 1, sophomore or 2, ...), and Major (mathematics or math, computer science or CS,);
- each COURSE record includes data to represent the CourseName, CourseNumber, CreditHours, and Department (the department that offers the course); and so on.

Database Manipulation

- Database manipulation involves querying and updating.
- Examples of queries are "retrieve the transcript-a list of all courses and grades-of Smith," "list the names of students who took the section of the Database course offered in fall 1999 and their grades in that section," and "what are the prerequisites of the Database course!"

What is a relational database?

- IBM defines a database as..
- ***A relational database is a type of database that organizes data into rows and columns, which collectively form a table where the data points are related to each other.***

Examples of Relational Database

Some of the most well-known RDBMSs include MySQL, PostgreSQL, MariaDB, Microsoft SQL Server, and Oracle Database.



A blurred photograph of an office interior. In the background, several people are seated at long wooden tables or desks, working. The scene is brightly lit, likely by large windows on the left, creating a high-key, airy atmosphere. The focus is soft, emphasizing the overall environment rather than individual details.

Job Opportunities

Database Focused Roles

- **Database Administrator (DBA):** Manage, secure, and optimize databases.
- **Database Developer:** Design and implement database structures, stored procedures, triggers, etc.
- **Data Engineer:** Build pipelines that move and transform data across systems.
- **SQL Developer:** Specialize in writing and optimizing SQL queries for applications.

Data Analytics and Science Roles

- **Data Analyst:** Query databases to create reports, dashboards, and insights.
- **Business Intelligence (BI) Developer:** Build reporting solutions using tools like Power BI or Tableau.
- **Data Scientist:** Use databases as the foundation for machine learning and predictive analytics.

Job Roles, Descriptions, Salaries

- Jobstreet.com
- Indeed.com

JobStreet.com

indeed